

School of Computer and Systems Sciences
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Database Management Systems (CS-706)
End Semester Exam, Monsoon Semester – 2023

Duration: 3 Hours

Max. Marks: 50

All questions are compulsory and carry equal marks:

Q1 (a). Consider the relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and set of FDs: $AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ$. Decompose relation R into 2NF and then in 3NF relations.

Q1 (b). Given two sets of FDs for a relational variable $R \{A, B, C, D, E\}$. Are they equivalent?

1. $A \rightarrow B, AB \rightarrow C, D \rightarrow AC, D \rightarrow E$

2. $A \rightarrow BC, D \rightarrow AE$.

Q2 (a). Add *lock* and *unlock* instructions to the following transactions $T1$ and $T2$, so that they observe the *two-phase locking* protocol.

$T1$: read(A), read (B), if $A = 0$ then $B=B+1$, write (B)

$T2$: read (B), read (A), if $B=0$ then $A=A+1$, write (A)

Q2(b). Consider a relation $R (A, B, C)$. Does the instance $R \{(a1, b1, c1), (a1, b1, c2), (a1, b2, c2), (a2, b1, c2)\}$ exhibit join dependency $JD [AB, AC, BC]$. If a tuple $(a2, b2, c1)$ is added to the instance R then how is join dependency $JD [AB, AC, BC]$ ensured?

Q3 (a). Consider the following schedule S : $T1$: Read(A), $T1$: Write(A), $T2$: Read(A), $T1$: Read(B), $T2$: Write(A), $T1$: Write(B), $T2$: Read(B), $T2$: Write(B). Show conflict instructions (if any) in the given schedule. Draw an equivalent conflict serializable schedule.

Q3 (b). Consider a relation $R (A, B, C, D, E, F)$ with MVDs $AC \twoheadrightarrow DF$ and $B \twoheadrightarrow E$. Decompose the relation R into 4NF relations.

Q4 (a). Determine the conflict equivalent serial schedule for the following schedule: $T1$: $R(X)$, $T2$: $R(Y)$, $T3$: $R(Y)$, $T2$: $W(Y)$, $T1$: $W(X)$, $T3$: $W(X)$, $T2$: $R(X)$, $T2$: $W(X)$

Q4 (b). Differentiate between *wait-die* and *wound-wait* deadlock preventions techniques.

Q5 (a). Suppose that we decompose the schema $R \{A, B, C, D, E\}$ into $R1\{A, B, C\}$ and $R2\{A, D, E\}$. Verify whether this decomposition is Lossy or Lossless if the functional dependencies $B \rightarrow D, A \rightarrow BC, E \rightarrow A$ and $CD \rightarrow E$ holds. Also, verify whether the decomposition is dependency preserving.

Q5 (b). Determine whether the schedule is valid under *Time-Stamp Ordering* protocol. Assume that $R-TS(A) = W-TS(A) = 0$, $R-TS(B) = W-TS(B) = 0$ and $TS(T1) = 5$ and $TS(T2) = 6$ for the following schedule:

$T1$: read(B), $T2$: read(B), $T2$: $B=B-50$, $T2$: write(B), $T1$: read(A), $T2$: read (A), $T1$: show ($A+B$), $T2$: $A=A+50$, $T2$: write (A), $T2$: show ($A+B$)